

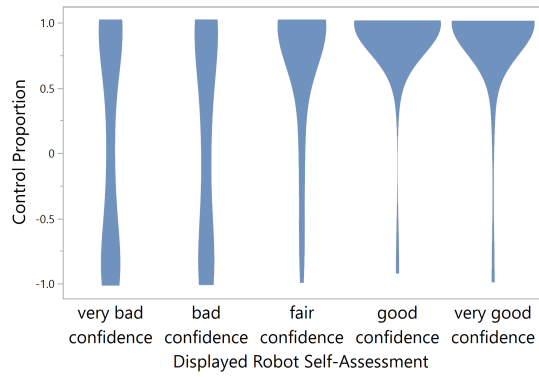


Figure 6: Box plot of control proportion for all robot self-assessment reports showing alignment between robot reports and control proportion. Positive control proportion indicated more robot control while negative control proportion indicated more participant control.

scores the importance of a robot being able to accurately self-assess. This data is shown in Fig. 6.

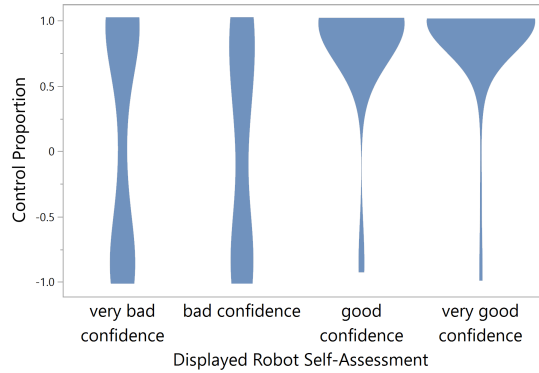


Figure 7: Box plot of control proportion for all informed robot self-assessment reports showing alignment between the robots reported confidence and control proportion. Positive control proportion indicated more robot control while negative control proportion indicated more participant control.

Informed self-assessment reports: We conducted a one-way analysis of variance (ANOVA) to test whether the presence of informed robot self-assessment reports affected the control proportion and found significant effect of informed reports on control proportion, $F(3, 29.04)$, $p < .0001$. Post-hoc comparisons using Tukeys HSD test revealed difference of control proportion between tasks reported as very bad and [good, very good] ($p < .0001$) and tasks reported as bad and [good, very good] ($p < .0001$). This data is shown in Fig. 7. Similar to the previous case, this indicated that participants were more likely to change to a level of autonomy appropriate for the displayed robot self-assessment report. However control proportion also shows us that Automatic Control mode was still used (though much less frequently) for tasks with poor self-assessments (very bad, bad). Further

exploration should investigate possible biases towards using the automation in the presence of robot self-assessments.

FUTURE WORK

We have shown the value of robot proficiency self-assessments to human-robot teams given simple short duration tasks. However, during more complex tasks or longer interactions an *a priori* self-assessment can quickly become stale, leading to the human inadvertently push a system beyond its competency bounds. Additionally, control proportion as described here assumed a fixed proficiency, and as such may be inappropriate for interactions in which the robots proficiency changes over time. In order capture the dynamic nature of realistic team interactions, our future work will focus on methods for real-time robot self-assessments and the development of performance metrics suited for online and longer term interactions.

CONCLUSIONS

Reporting of a robot's *a priori* proficiency self-assessment can provide valuable information to improve a human teammate's decision making process. In this study we found that the reporting of robot self-assessment, in the form of rewards based FaMSeC Outcome Assessment, resulted in a decrease in task failures and small improvements to the participants self-reported trust. An *a priori* robot self-assessment essentially aligns the participants task expectations with the robots belief in its abilities. Our objective trust metric, control proportion, gave us insight into the participants choice of autonomy allocation: tasks in which the robot reports lower self-confidence in meeting user expectations showed an increase in participant control, while tasks in which the robot reported higher self-confidence in meeting user expectations showed an increase in autonomous control. This work is an incremental step in improving trust and performance within a human-robot team.

Acknowledgments

The authors would like to thank Dr. Brett Israelsen (Raytheon Technologies Research Center) for his help reviewing this document, and for the many informative discussions relating to trust, machine self-confidence, and human subjects studies.

Ipsam placeat enim ullam omnis laborum, voluptates earum esse assumenda magnam velit, sapiente illo porro beatae harum qui magnam optio ad temporibus quibusdam ea, sed praesentium facilis. Aperiam similique beatae quis at enim, atque maxime adipisci vitae dolorum ullam quidem architecto, nesciunt dolores accusantium tempore corrupti distinctio delectus harum qui. Id nemo molestiae non assumenda magni quas praesentium quisquam dolor cupiditate reprehenderit, expedita vero quaerat nobis beatae optio, excepturi rerum eveniet animi officiis quidem nemo, adipisci officia amet quo ipsam corrupti asperiores placeat atque fugit. Suscipit facilis eius a veniam, perspiciatis fugiat obcaecati soluta consequuntur doloremque sapiente eveniet quasi nulla atque nam, quis molestiae consequatur rem. Perferendis ut aperiam, temporibus iste of-

ficia nam, debitis accusantium animi nulla placeat itaque
beatae, voluptatibus maiores qui.Ea in laudantium eum, fu-
giat ducimus quam sint amet, corporis cumque